

SHUBHANGI VATS

B.TECH (IOT), 2ND YEAR
MINOR: SPACE TECHNOLOGY

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EDUCATION

B.Tech in Internet of Things

Thakur College of Engineering
2024 – 2028

Higher Secondary (XI–XII)

Abhinav English School(CBSE), Pune

TECHNICAL SKILLS

- **Programming:** Java, C (Basics)
- **Embedded & IoT Systems:** Arduino, ESP32, Vega IoT Boards (CDAC)
- **Tools & Platforms:** LabVIEW, AWS (Cloud Fundamentals)
- **Core Domains:** Embedded Systems, Sensor Integration, IoT Architecture
- **Specialization:** Scientific Observations from Space (START)

INVOLVEMENT

- **Astronomy Club (AC-TCET)**
Upcoming Documentation Head (2026–2027)
- **Astronomy Club (AC-TCET)**
Documentation Committee Member (2025–2026)
- **Winter Intern**
Cyber Peace Foundation (2024–2025)
- Participated in **Student Exchange Programs** on Carbon Footprints, Climate Change, and Cultural Exchange.
- Undertaking a **one-month Technology Internship** in **Dubai** from mid-June 2026.

PROFILE

Research-driven student with a strong focus on space technology, embedded systems, and intelligent sensing applications. Hands-on experience in developing IoT-based prototypes, safety systems, and conceptual aerospace solutions. Passionate about solving real-world challenges through interdisciplinary engineering, with proven ability to work in structured, high-performance environments.

PROJECTS

FIRE DETECTION & AUTO-RESPONSE SYSTEM

- Developed a sensor-based fire detection system using flame and temperature sensors (Arduino-based prototype)
- Reduced response time from ~10 seconds (manual detection) to ~4 seconds (automated trigger)
- Achieved ~92% detection accuracy across 25+ controlled test trials
- Integrated automated extinguishing mechanism for rapid hazard mitigation

IOT-BASED INDUSTRIAL WORKER SAFETY HELMET

- Built a smart safety helmet using Vega IoT board with gas and motion sensors
- Enabled real-time monitoring with alert latency under ~2 seconds
- Tested across 20+ simulations, improving incident response readiness by ~30–35%
- Designed for scalable deployment in industrial safety environments

DISASTER RESPONSE DRONE (SMART INDIA HACKATHON)

- Designed a conceptual drone system for emergency response in remote terrains (Odisha use case)
- Estimated ~25–30% reduction in response reach time compared to traditional ground methods
- Proposed payload system for medical supply delivery up to 2–3 kg (concept design)
- Focused on terrain adaptability and rapid deployment architecture

RESEARCH

- Authored **"Safeguarding Networks from Hackers and Attacks in India,"** for Multicon analyzing 12+ network attack vectors (including phishing, DDoS, and MITM) and proposing multi-layered defense strategies that improved theoretical threat mitigation coverage by ~40%.
- Published **"The 'Brain' in the Satellite: How Sensor Fusion is Powering Smarter Space Systems"** in ThinkTech IoT Magazine (2025), exploring sensor fusion, AI-driven satellite autonomy, and IoT-integrated space technologie